Improving Quadruped Stabilization with a Tendon-Driven Segmented Tail

Kausik Kolluri, Atharva Ramdas, Aryan Chandra, Raymond Cao, Jash Lapsiwala

[1]

***Abstract*— This project investigates the use of an actively actuated and modular segmented tail to improve the stability of quadrupedal robot locomotion. Motivated by biological examples in which animals use flexible tails for balance and disturbance rejection, we propose a base-actuated segmented tail integrated into an adapted SpotMicro quadruped platform. The system is evaluated through controlled perturbation experiments and straight-line walking trials, with stability quantified using peak roll and yaw deviations measured via onboard IMU data. Additionally, the effect of tail mass distribution on stabilization performance has been studied through three distributions: uniform, front heavy and rear heavy while maintaining a constant total tail mass. Through the level walking experiments it was observed that an active tail demonstrates overall lower peak yaw angles compared to an inactive tail, with the rear heavy tail type stabilizing the quadruped to the lowest peak yaw angles overall. Results from the perturbation experiments were restricted to below 10° due to the quadruped's standing position, revealing no distinct different between the active and inactive tails.**

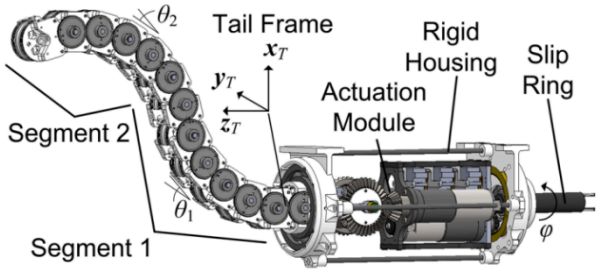
I. INTRODUCTION

Quadrupedal robots have been a promising focus in the robotics community for the last half century, beginning with dog-like robots meant for rough terrain navigation from the CMU Leg Laboratory and the MIT LegLab in the 1980's [1]. With the development of bioinspired and biomimetic designs, engineers soon progressed towards legged robots that adapt to changing terrain while performing autonomous tasks, including leakage detection and safety inspections [2], [3]. Though these advancements have propelled the area of quadrupedal robots forward, a broad problem which has persisted while navigating rugged terrain is maintaining balance and stability. Stabilization and maneuvering of quadrupedal robots demand systems with at least twelve degrees of freedom, involving sophisticated algorithms that must induce quick responses to environmental stimuli [4], [5]. While different groups approach this issue with unique methods such as complex controllers and motion planning algorithms, some engineers are leaning towards stabilization methods employed by animals in nature for inspiration [5], [6].

Across many species of animals, long, flexible tails provide full body control and maneuverability. For example, lizards swing their tails during aerial maneuvers for reorientation and self-correction purposes, while kangaroos use their tails as a counterbalance when standing and for energy storage during hopping motions [7], [8]. Motivated by these biological strategies, engineers addressed the issue of quadruped stability in variable terrain by designing, fabricating, and testing counterbalancing tails. Early robotic implementations simplified the tails of lizards and small cats to a 1 degree of freedom (DOF) model, as demonstrated in an article published by Zhang et al. in 2016. The group demonstrated that tail swinging in roll reduced yaw errors, enabling the robot to maintain its direction when trotting [8]. To address the planar limitations of a 1-DOF tail, Yang et al. designed a 2-DOF tail to improve contact-independent control of a quadruped robot meant to traverse extreme terrain. The 2-DOF tail proved to reduce maximum angular error while climbing a step of varying heights [9].

While these examples provide a unique insight into the benefits of tail stabilization, animal tails are rarely rigid in nature with continuously varying curvature and inertia increasing degrees of freedom and enhancing dynamic control. For this reason, Rone et al. developed and experimentally validated a multisegmented serpentine tail capable of actively stabilizing and maneuvering a quadrupedal robot during locomotion [4]. Collectively, these results highlight the potential of tail assisted stabilization. However, gaps remain in the literature on stabilizing quadrupeds using tails:

1. Current segmented tails on quadrupeds have a highly complex mechanical design: Literature shows the mechanical design complexity increasing with the number of degrees of freedom of tails. For instance, the segmented tail in [4] consists of an actuation module with multiple levels of gearing required and three required control inputs to move the tail, as shown in Figure 1. Simplifying the tail design while



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maintaining its high dexterity remains a key challenge in achieving animal-like tail functionality.

Figure 1. High complexity of the twelve DOF tail actuated by 3 control angle inputs. [4]

2. There have not been studies on mass distribution along robotic tails: Work on the effect of different tail lengths on the body pitch per stride, performed by Heim et al using a 1 DOF tail, showed that long and light tails are more effective than short and heavy tails with the same moment of inertia [10]. However, their tails were designed as a lightweight rod with a mass at the end, and not as a continuous mass distribution. It therefore remains an open question on how tail mass distribution influences the control effort required for tail actuation.

Taking inspiration from the clouded leopard – which uses its long, slender tail to counterbalance instability during arboreal motion – we designed a modular, segmented tail actuated with tendon driven actuation [11]. This allowed us to address these gaps by mimicking the swaying and curling motion of a clouded leopard tail using a two-motor pulley system rather than a system of gears. By calculating peak yaw and roll angles of the quadruped during level walking of the quadruped, we were able to answer our first research question: **"Does a segmented tail actuated only at the quadruped base reduce peak yaw and roll angles during walking and perturbations compared to a rigid unactuated tail?"** Furthermore, with the segmented design facilitating changes in mass distribution we were able to understand the effects on counterbalancing: **"How does the mass distribution through a segmented tail actuated at the base of a quadruped affect peak yaw and roll angles?"** This report demonstrates the design, fabrication, and control of a tail stabilized quadruped, along with the results from the two experiments conducted to answer the proposed research questions.

II. QUADRUPED AND TAIL DESIGN

A. Quadruped Design

The preliminary design is based on a 12-DoF open-source SpotMicro quadrupedal robot platform with CAD as shown in Figure 2. The redesigned quadruped torso introduces dual-platform architecture to improve component organization, expand hardware compatibility, and simplify wiring management. Unlike the original SpotMicro torso, which has limited internal volume and is primarily constrained to a single-board configuration of NVIDIA Jetson Nano, this design separates computing and control hardware across two levels (Supplementary Figure D).

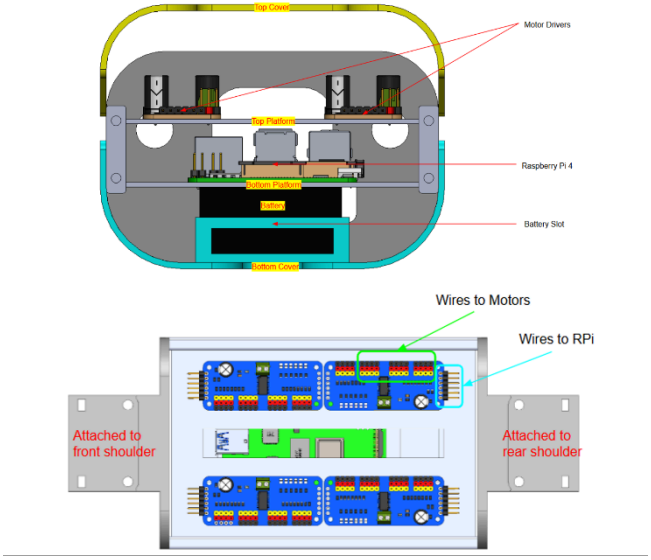
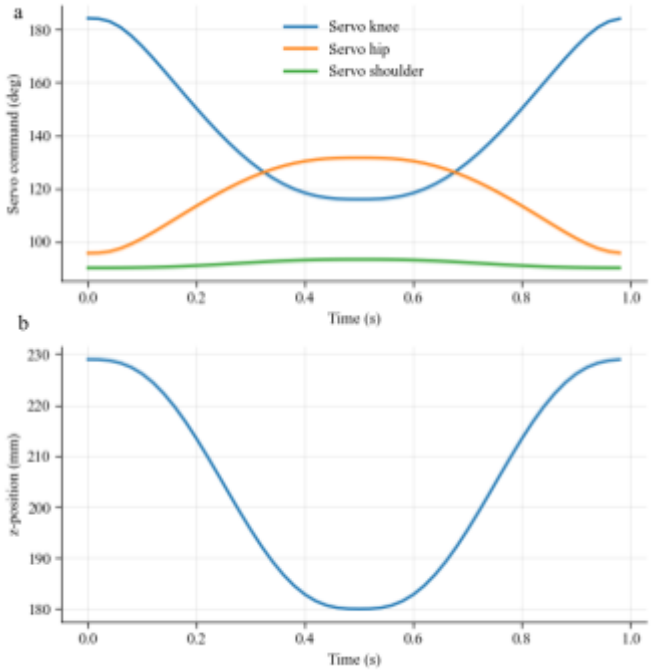


Figure 2. Redesigned quadruped torso side view (top) and bird eye view (bottom) without covers.

The lower platform is dedicated to housing a Raspberry Pi, providing a stable and accessible location for the primary processing unit. Positioning the Raspberry Pi on the bottom layer improves accessibility for ports and allows for better cable routing towards peripherals and external connections. The upper platform is reserved for motor drivers and power electronics. By isolating these components from the main computing board, the design reduces clutter, minimizes electrical interference, and improves thermal distribution. The elevated placement also allows for shorter, more direct connections to the motors mounted on the quadruped's limbs.

To further enhance system integration, the bottom cover incorporates a dedicated slot for battery placement. This feature secures the battery within the torso while maintaining a low center of gravity, which is beneficial for overall stability during locomotion. To address wiring complexity, the torso incorporates strategically placed holes and slots throughout both platforms. These features enable organized cable routing between layers, reduce wire congestion, and improve maintainability during assembly and troubleshooting.



B. Inverse Kinematics

Procedure: The inverse kinematics for each leg is calculated independently, by fixing an identical coordinate system on each shoulder with origin at the center of the shoulder joint, the x-axis pointing forward, the z-axis vertically down as shown in Supplementary Figure A. This enables commanding each leg with the same trajectory (with time/phase offset between diagonal pairs in the case of walking gait). This technique solves for all 3 degrees of freedom of a spot-micro leg, by separating the problem of obtaining joint angles from a foot position into 2 planes- the plane facing the front of the robot and the plane containing the knee and hip, because two of the joints (knee, hip) move in the same plane and the third joint (shoulder) moves in a perpendicular plane. This implementation solves for a single solution that assumes all joint angles have a range motion of 180 degrees, which is within the angle range required for a walking gait, therefore helping avoid redundant solutions.

Validation: Calculations are validated through line diagrams that help visualize how the leg looks from both the front plane and the hip-knee plane. Figure 3 shows such diagrams intended for the left leg, one with a straight leg configuration and the other with the leg displaced (by 40mm in negative z-direction) vertically from the initial configuration, leading to a more folded leg configuration. Next, the legs are provided with a polynomial position trajectory to move vertically up and back down, and the resulting joint angle and servo angle trajectories calculated from the inverse kinematics are plotted against time and inspected to confirm feasible servo trajectories as per our servo zeroing convention, as shown in Figure 4.

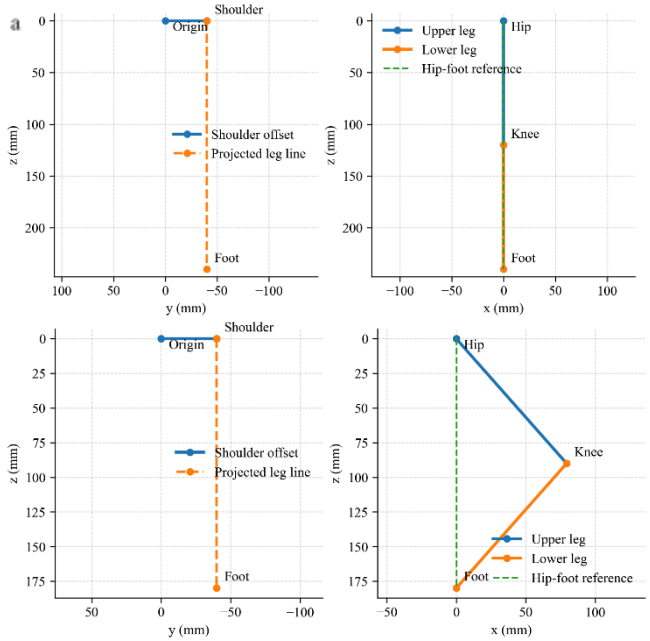


Figure 3. a) Line diagram for straight leg configuration, input foot z-position = 240 mm. b) Line diagram for folded leg configuration, input foot z-position = 180mm (-60 mm z displacement).

Once the inverse kinematics was implemented on the physical robot, its legs were commanded to move along the vertical axis (z-axis). To measure leg movements, a test rig was used where the legs were suspended in the air, and a ruler was attached to the stand with its zero-mark aligned horizontally with the leg's hip joint. The leg was then commanded to move vertically between z = 18.5 cm and z = 23.5 cm, and the actual positions were recorded as the leg passed along the ruler. The same commands were applied to a simulated Spot Micro leg on Mujoco, and its foot positions were extracted over time into a CSV file for comparison. The videos of inverse kinematics tests on simulation and the physical robot can be found in Supplementary Videos A and B, respectively, and the video of all four legs moving together can be found in Supplementary Video C. In addition, an image analysis of the movements to demonstrate vertical leg movement, can be found in Figure 5.

Figure 4. a) Fifth order polynomial position trajectory for vertical motion. b) Corresponding servo trajectories from inverse kinematics calculations show that all servo commands calculated lie within servo 270 degree motion range

Following the verification of the inverse kinematics, a keyframe-based trotting gait was implemented for the robot, as shown in Supplementary Video D. The stance phase consists of a linear trajectory and the swing phase consists of a semicircular trajectory, as described by Figure 5. For the stance phase:

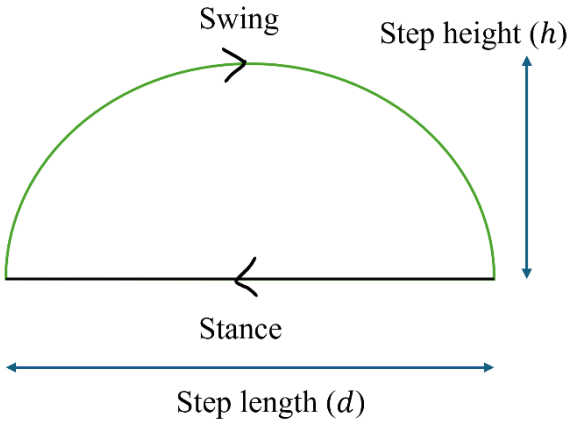


Figure 5. Foot trajectory for trotting

$$x = x_0 + \frac{d}{2} - sd \quad (1)$$

where x_0 is the standing x-position of the foot, d is the step length, and s is the fraction of the stance duration that has passed, leading to the leg moving linearly from

$$x = x_0 + \frac{d}{2} \quad (2)$$

to

$$x = x_0 - \frac{d}{2} \quad (3)$$

through the stance phase. The y and z position is constant (=y0, z0, the standing positions) for the stance phase. For the swing phase:

$$x = x_0 + \frac{d}{2}\cos cos\, s\pi \; - \; sd$$
$$z = z_0 + h \sin sin\, s\pi \; - \; sd$$
$$y = y_0 \qquad (6)$$

where h is the step height, leading to the leg moving in a semicircular trajectory from

$$(x_0 - \frac{d}{2},\, z_0) \qquad (7)$$

to

$$(x_0 + \frac{d}{2},\, z_0) \qquad (8)$$

with z first increasing then decreasing.

C. Tail Design

The tail design had two primary objectives: achieving a curling effect with one degree of freedom (1DOF), and accommodating variable mass distribution along the tail's length. To meet these goals, we designed a segmented tail, with each segment having a square cross-section of uniform width.

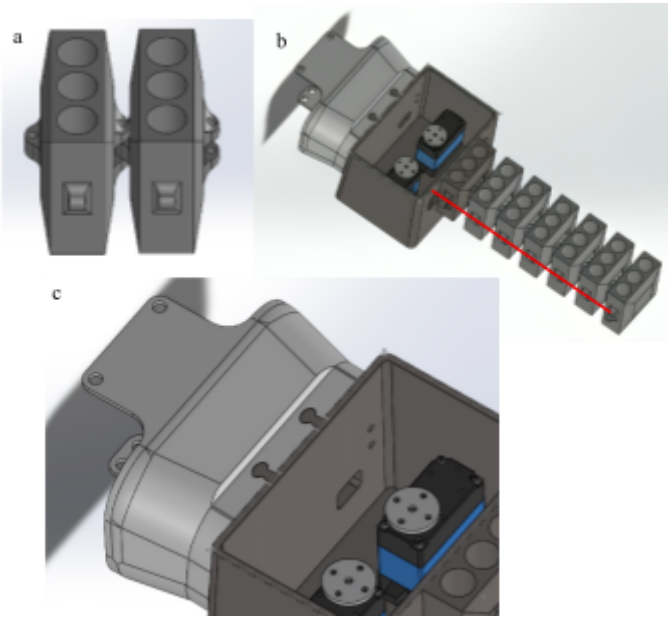


Figure 6. a) Knuckle-type joint that allows constrained rotation with 1DOF. b) Full tail assembly showing the position of the motor with tendon shown in red. c) The sliding dovetail joint between the tail and the quadruped rear cover

Each segment incorporated two important design features that helped us achieve both objectives. One face of each segment had a fork end, while the opposite face had a tongue joint, allowing adjacent segments to be connected using a 2mm bolt (pin). This knuckle joint constrains motion to 1DOF rotation, producing the intended curling effect. To enable mass distribution along the tail, each segment was designed with three cylindrical cavities capable of housing upto 12 bearing balls (0.5in. diameter, 8.5g each). The cavities were sized for a clearance fit, ensuring the balls remain free to settle without contributing nonlinear inertial effects during tail actuation.

To actuate the tail, each segment included small hooks on its side faces for tendons routing. The frontmost segment featured only the tongue joint, and the tendons were anchored at its side hooks. The other ends of both tendons were connected to two pulleys attached to a servo motor, tensioned such that the tail remained straight at the motor's zero position. To enable actuation in both directions, the motor's zero position was set at the midpoint of its angular range, at 135°. For ease of testing, the ail motors were fixed at one end within a dedicated mounting box. To allow quick attachment and removal, the rear cover of the quadruped and the motor mount box were connected using a clearance fit sliding dovetail joint, thereby constraining the tail assembly against lateral movement.

D. PID Tail Control

To drive an active tail, we implemented a PID controller to track the error in either the yaw or roll angles, depending on the experiments we performed. A yaw controller was used for the level walking experiment, while a roll controller was used to stabilize the quadruped during the perturbation test. Defining θ_{yaw} and θ_{roll} as the real time yaw and roll angles read by the IMU, equations for tail actuation were created

$$\alpha = \theta_{ref} - \theta_{yaw} \qquad (9)$$
$$\beta = \theta_{ref} - \theta_{roll} \qquad (10)$$

where α and β, are the angles the tail must cover to stabilize the quadruped for the yaw and roll perturbations, respectively, and θ_{ref} is the respective reference angle (0° for both yaw and roll in the experiments). These equations were then related to the motor angle rotations δ_{yaw} and δ_{roll}

$$\delta_{yaw} = K_p\alpha + K_d\dot{\alpha} \qquad (11)$$
$$\delta_{roll} = K_p\beta + K_d\dot{\beta} \qquad (12)$$

where K_p and K_d are the proportional and derivative gain constants used to control tail swing speed and oscillations, respectively.

A PD controller was used for yaw and a P-only controller for roll. The derivative term in the yaw controller was intentionally kept small relative to the proportional gain, allowing the tail to carry slight momentum and impart a corrective impulse to bring the quadruped back to its desired trajectory. Before each test, the quadruped was placed in its desired orientation and the initial yaw and roll angles were recorded as the reference, effectively zeroing the error at the start of each experiment.

Both controllers output a target angle for the motors, which curl the tail in the desired direction. The tendon and motor configuration was arranged such that both motors receive the same input angle to produce curling in either direction.

III. EXPERIMENTAL METHODS

A. Level Walking

The first experiment focused on observing the quadruped during short distance, level walking. Tape was used to mark the starting and ending positions of the walk and were chosen based on an average walking time of 45 seconds. This time was chosen to result in a consistent number of peak yaw angles within the selected time frame of 10 seconds and 20 seconds, avoiding noisy peaks found at the beginning and end of each walking trial. The control experiment observed the walking gait of a rigid, inactive tail which was held straight along the quadruped midline. Peak yaw angles we extracted from 3 walking trials for the inactive and active tails with 3 different mass distributions being observed – front heavy, rear heavy, and uniform – resulting in 18 total trials. The uniform mass distribution was defined by 1 ball bearing placed in the center slot of each of the 6 segments. The front-heavy distribution had all 6 ball bearings in the moveable segment closest to the quadruped torso, while the rear-heavy distribution had all 6 ball bearings placed in the last segment furthest away from the torso. The peak yaw angles were plotted as box and whisker plots for each trial and then combined for each group (ex. Front heavy, inactive tail). Statistical analysis consisted of a normality test, followed by ANOVA testing, and finally a pairwise t-test comparing the 3 mass distribution groups for an inactive and active tail separately. The code for extracting peak yaw angles and statistical analysis can be found in the Supplementary Material Section 2.

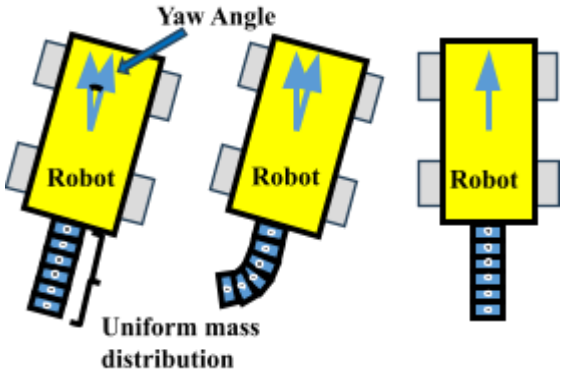


Figure 7. Diagram of level walking example with uniformly distributed ball bearing mass with defining yaw angle

B. Pendulum-Driven Perturbation

The second experiment focused on perturbing the quadruped along its roll axis using a pendulum to transfer momentum above the front legs of the quadruped. The quadruped starting position was marked by tape to keep the impact spot consistent. By using a pendulum dropped at similar heights, consistent momentum transfer was maintained during each trial. Similar to the walking experiment, 3 trials were performed for 3 mass distributions on inactive and active tails, in which each trail consisted of 3 impacts with a 10 second delay. The 3 peak roll angles for each trial were adjusted by subtracting the peak height from the baseline, which changed within each trial. The peak roll angles for inactive and active tail were compared for each mass distribution as box and whisker plots with the 9 peak angles being plotted as scatter points as well. Furthermore, recovery times were defined by the time between impact and when the roll angle stabilizes to within 10% of the baseline after impact. Statistical analysis consisted of a normality test, followed by ANOVA testing, and finally a t-test comparing the active and inactive tails for each mass distribution separately. The code for calculating the peak roll angles and the recovery times can be found in the Supplementary Material Section 2.

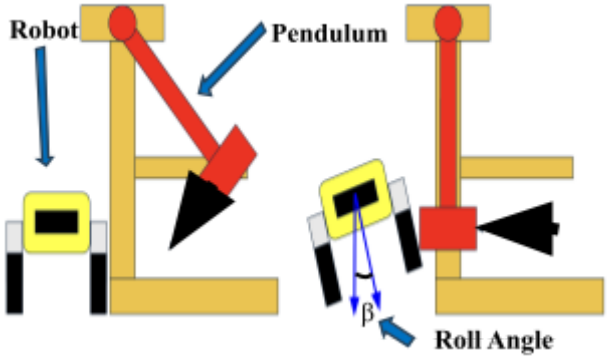


Figure 8. Diagram of pendulum-driven perturbation defining roll angle.

III. EXPERIMENTAL RESULTS

A. Level Walking

The level walking results show that the actuated tail reduced average peak yaw angles across all three mass distributions compared to the unactuated rigid tail. In the unactuated case, the statistical comparisons show significant differences between mass distributions, with several pairwise comparisons reaching strong significance, including $p<0.017$ after Bonferroni correction and $p<0.001$. This suggests that when the tail is passive, mass distribution has a measurable effect on yaw instability. In contrast, the actuated tail condition shows generally lower peak yaw angles and reduced spread across front-heavy, rear-heavy, and uniform cases, indicating improved yaw regulation. The significant p-values support that the observed differences are unlikely to be due to random variation alone, and the lower yaw magnitudes in the active-tail trials suggest that reactive tail control improves walking stability by suppressing excessive body yaw across all tested mass distributions.

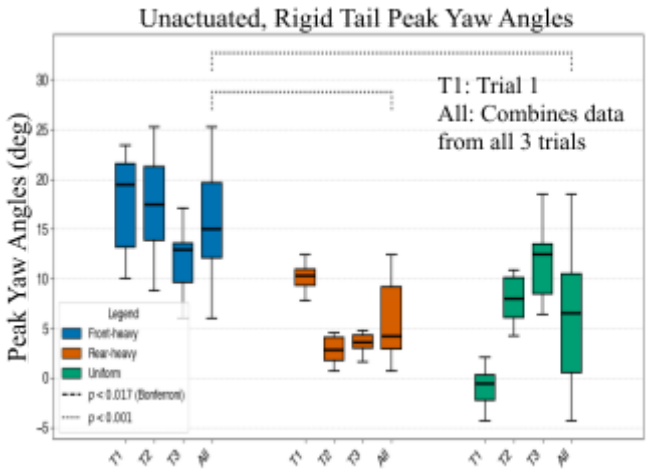


Figure 9. Peak yaw angle data for an unactuated, rigid tail during level walking. Mass distributions shown in order from left to right; front heavy, rear heavy, uniform. All 3 trials shown for each mass distribution along with the cumulative data shown in the "All" box plot.

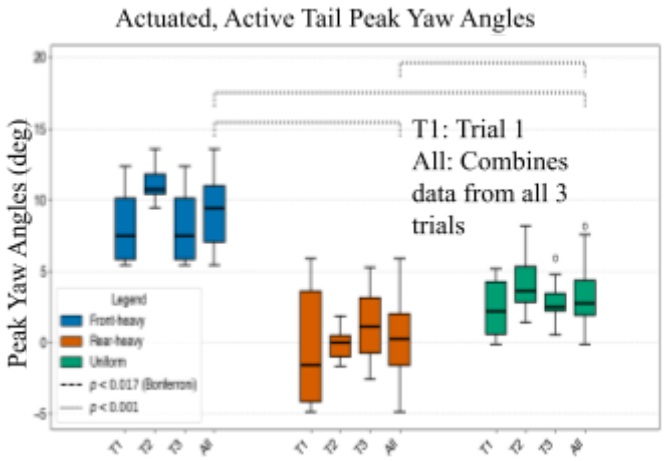


Figure 10. Peak yaw angle data for an actuated, active tail during level walking. Mass distributions shown in order from left to right; front heavy, rear heavy, uniform. All 3 trials shown for each mass distribution along with the cumulative data shown in the "All" box plot.

B. Pendulum-Driven Perturbation

The peak roll angle results from the pendulum-driven perturbation experiments are presented in Figure ___. Since the objective of this study was to compare the stabilizing effects of an actuated tail and an unactuated tail, the data were grouped by tail mass distribution. Following the statistical analysis procedure described in the Methods section, we obtained p-values greater than 0.05 for all three mass distributions when comparing active and inactive tail configurations. These results indicate that there is no statistically significant difference in peak roll angle between actuated and passive tails across any of the tested mass distributions. Analysis of perturbation recovery time further supports this conclusion, as no significant differences were observed between tail types (Supplementary Figure B).

Collectively, these findings suggest that for roll angles below 10°, a passive, rigid tail provides roll stabilization performance comparable to that of an actively actuated tail. This result is likely due to the quadruped's rigid leg structure in the standing position. In this configuration, the torque provided by all three leg motors prevent larger roll angles, allowing the restoring torque generated by the robot's weight to dominate the roll stabilization response.

Comparisons across tail mass distributions similarly reveal no statistically significant differences in peak roll angle. Although the front-heavy unactuated tail exhibits the smallest variability, this behavior can be explained by the shortened distance between tail mass to the robot's overall center of mass. A more localized center of mass will reduce the moment arm and limit roll excursions. Therefore, for roll angles below 10°, peak roll magnitude and variability in a stationary quadruped are largely insensitive to tail mass distribution.

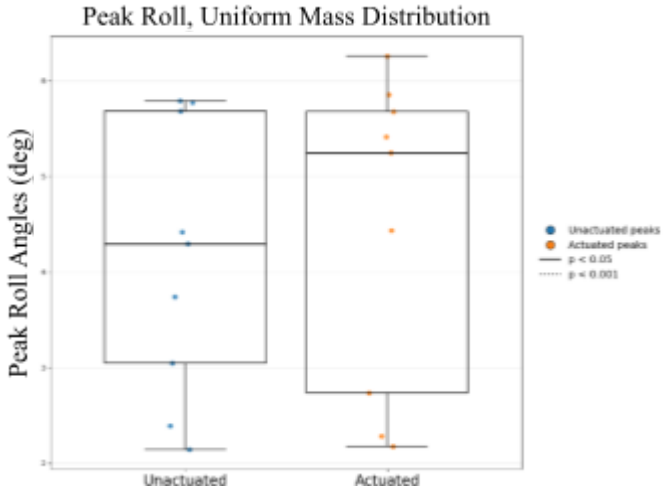


Figure 11. Peak roll angles for the uniform mass distributed tail during perturbation experiments

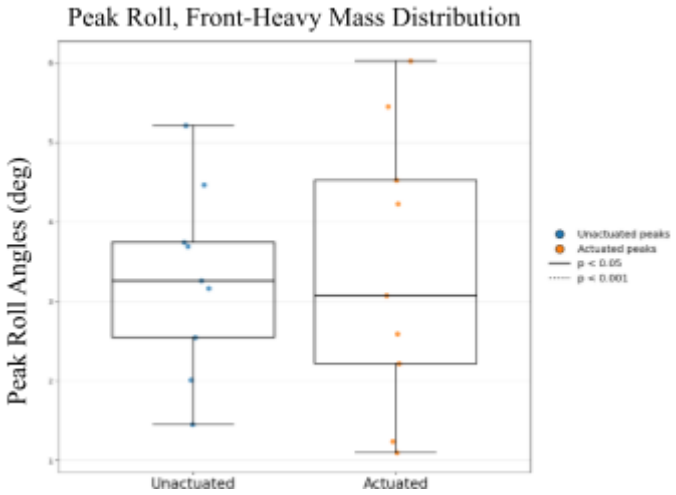


Figure 12. Peak roll angles for the front-heavy mass distributed tail during perturbation experiments

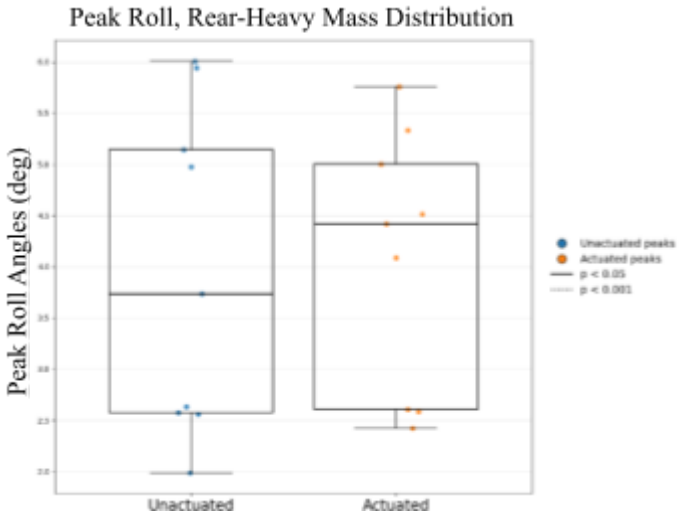


Figure 13. Peak roll angles for the rear-heavy mass distributed tail during perturbation experiments

V. Conclusion and Future Work

This report details the design, fabrication, and testing of a tail stabilized quadruped. Modeled after the SpotMicro, the quadruped features a segmented tail of length proportional to its torso with tendon driven actuation and an integrated PID controller focused on yaw and roll angle deviations in the quadruped. Level walking and pendulum driven perturbation experiments were conducted to test the yaw and roll stabilization capabilities of the tailed quadruped, respectively. Three different tail mass distributions we also tested as a part of each experiment to understand if there is an effect on tail-driven stabilization. The yaw and roll angles of the quadruped during the experiments were extracted and analyzed. Level walking experiments demonstrated lower peak yaw angles when comparing an actively, actuated tail with a rigid, unactuated tail. The rear-heavy actuated tail performed with the lowest peak yaw angles. The perturbation experiments yielded peak roll angles below 10° due to the standing position of the robot leading to gravity becoming the primary restoration force rather than tail actuation. Therefore, across the three tail mass distributions, the unactuated and actuated tails performed similarly. As a part of future work, we plan to design a more force-resistant casing for the robot to prevent damage during perturbation testing. This will allow for larger roll angles to demonstrate a possible difference between unactuated and actuated tails.

Appendix

Section 1. Supplementary Figures and Videos

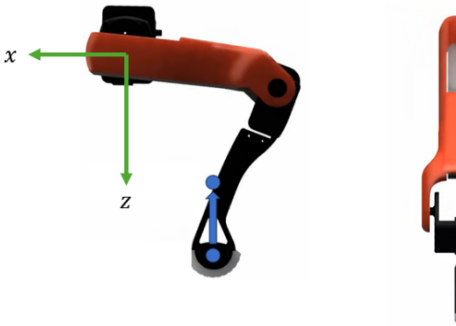


Figure A. Coordinate frame setup for left leg: back view (left) and size view (right). Images obtained from: Inverse Kinematics for SpotMicro robotics | example and demo, *Advanced Hobby Lab*. Youtube

	Mass Distributions					
Tail Types	Front Heavy		Rear Heavy		Uniform	
	Recovery Times (s)	Average ± Std. Dev (s)	Recovery Times (s)	Average ± Std. Dev (s)	Recovery Times (s)	Average ± Std. Dev (s)
Unactuated	1.29 0.87 1.14 0.88 1.10 1.06 0.73 1.03 0.64	0.97 ± 0.21	0.82 0.68 1.01 1.00 0.88 1.06 0.80 0.86 0.75	0.87±0.13	0.61 0.63 0.77 0.88 1.08 0.61 0.89 0.88 1.03	0.82±0.18
Actuated	0.87 0.91 0.75 1.01 0.91 0.87 0.70 0.33 0.93	0.81±0.20	0.98 0.85 0.79 0.90 1.00 1.54 0.88 0.98 1.08	1.00±0.22	0.79 0.60 1.08 0.84 0.48 0.58 0.96 0.62 1.20	0.79±0.25

Figure B. Table of recovery times separated by tail type and mass distribution. Each trial is separated in each cell and the average and st. dev. is calculated from the 9 values from the 3 trials

Video A: inverse kinematics test on simulation
Video B: inverse kinematics test on physical robot
Video C: four-legged movement of robot based on inverse kinematics
Video D: walking gait test

Section 2: GitHub Repository and Google Drive

All CAD files, code and experimental videos for this study can be found in this GitHub repository.

The code used to analyze data and generate plots can be found in this Google Drive Folder.

Section 3. Individual Technical Contributions

Kausik: I was primarily responsible for the dynamic analysis and motor selection that enabled the quadruped's tail-based stabilization system. Converting my prior experience from kinematics and dynamics physics course into practical robotics

experience, I performed detailed dynamic calculations to determine the motor torque requirements necessary for the quadruped to remain within its support polygon while bearing the added mass of the tail. I independently derived and analyzed the forces and moments acting on the robot during perturbations to ensure structural viability and stability. In addition, I conducted dynamic calculations for tail swing velocity and angular range to minimize recovery time following roll perturbations. These calculations directly informed the selection of a motor capable of achieving the desired tail motion while operating safely below stall torque limits. I validated if the chosen motor could deliver sufficient torque and speed to meet the system's performance requirements without risking motor saturation or mechanical failure. Following the data processing, I independently organized and plotted the data while performing statistical analysis including the normality tests, ANOVA tests, and t-tests for all available data sets.

This work had a significant impact on the overall project, as it ensured that the tail actuation system was both feasible and appropriately understood before integration and testing. My analysis directly supported the experimental objectives outlined in the proposal and enabled reliable experimental comparisons between active and passive tail configurations. These tasks were completed independently, and the resulting calculations and motor specifications were adopted by the team without requiring substantial revision.

Atharva: I was responsible for developing the walking gait for the quadruped robot. This involved deriving the leg inverse kinematic equations, testing them using line-diagram visualizations, and experimentally verifying the calculations on the robot. I then researched typical quadruped gait patterns and deployed a trotting gait, where diagonal pairs of legs followed semi-circular trajectories with phase offsets. All core software for the quadruped was written by me, while my teammates assisted with inverse-kinematics tuning to maintain balance and improve straight-line motion. Overall, my subsystem was completed successfully and met the requirements for the planned experiments.

Aryan: I was responsible for locomotion development, actuator validation in sim, kinematic modeling, perturbation system design, and IMU data processing for the SpotMicro quadruped. I independently designed and implemented a keyframe-based trotting gait in MuJoCo using diagonal leg pairing, which was later refined with a semicircular swing phase to improve stability and foot trajectory. I validated this gait through simulation and logged per-joint torques across the gait cycle, directly informing actuator selection by ensuring all motors operated within safe limits. This early validation significantly reduced hardware risk and enabled confident system design before physical deployment.

I also developed a Python-based inverse kinematics (IK) solver that converted desired foot positions into joint angles, which was used during both simulation and hardware bring-up. To support robustness testing, I designed and built a perturbation rig with a pendulum-based impact system and created a physics model to estimate the momentum imparted to the robot. Although automated triggering was limited by trial variability, the system was successfully adapted for manual use in experiments.

Finally, I independently developed the IMU data processing pipeline used for all experimental analysis. This included cleaning noisy raw sensor data, correcting orientation inconsistencies, and generating structured outputs for visualization and statistical evaluation of roll, pitch, and yaw. This pipeline enabled reliable interpretation of results and directly supported the project's experimental conclusions.

Overall, my contributions were completed with minimal external assistance and formed a critical foundation across simulation, hardware validation, and data analysis, ensuring the project met its technical and experimental objectives.

Jash: I was responsible for the mechanical design of the tail, including the segment design ideation, geometry selection and overall tail assembly. This also involved multiple iterations of rapid prototyping and testing, with all parts fabricated on my personal 3D Printer. I also designed the tendon driven actuation system of the tail, and developed the roll and yaw based PID controllers for experimentation. All the PID controller code for the tail actuation was written by me, while my teammates helped me tune the PID gain values. Apart from these, I led the fabrication of all the structural components of the quadruped, handling the full manufacturing workflow, including DfM analysis, part slicing and 3D Printing.

Raymond: My primary contribution to the project focused on the iterative mechanical and structural design of the quadruped robot, with an emphasis on integrating updated hardware components and improving overall system organization. I was responsible for modifying the leg design to accommodate newly selected motors. This required careful consideration of motor dimensions, mounting interfaces, and load distribution to ensure both compatibility and reliable performance without compromising the robot's mobility. In addition to the leg redesign, I developed a dual-platform layout within the robot's torso to efficiently organize and house all electronic components, including motor drivers, the Raspberry Pi board, the battery, and associated wiring. This layout improved accessibility, reduced clutter, and enhanced thermal and spatial management compared to the initial single-layer configuration. A key aspect of my contribution was continuous design iteration in response to integration challenges encountered during assembly. For instance, when spatial constraints prevented proper routing of wires between the battery, motor drivers, and the Raspberry Pi, I identified and implemented solutions such as elevating the quadruped's top cover to create additional internal volume. These adjustments ensured that wiring could be routed safely without excessive bending or interference with moving components.

Section 4: Budget

Active budget tracking link can be found here: **Project Budget Tracker**

Section 5: AI Usage Statement

We utilized AI tools (ChatGPT, Claude) to assist with grammar correction, coding and minor sentence refinement throughout the report. For writing, we provided sections of our drafted content and requested revisions to improve clarity, grammar, and overall readability. For coding, the AI agent assisted with creating well-commented code, and making plots fit the correct requirements of the assignment.

The use of AI was limited to language enhancement, and programming our ideas, and did not contribute to the technical content or design decisions.

References and Footnotes

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